

Problem Statement:

The Global Superstore dataset contains a large volume of sales transactions with information about customers, products, regions, dates, sales, quantity, discount, and profit. Analyzing such data directly can make it difficult to identify trends, compare business performance, and obtain meaningful insights from different perspectives.

The problem is to implement Online Analytical Processing (OLAP) operations using Microsoft Power BI by organizing the Global Superstore data into a suitable dimensional model. The model should support analysis across different dimensions such as time, product, customer, and region, while providing measures such as Sales, Profit, Profit Margin, and Average Order Value.

The solution should enable users to perform OLAP operations such as Slice, Dice, Drill-down, and Drill-up/Hierarchy analysis through Power BI visualizations, slicers, and interactive reports. This will allow users to examine specific subsets of data, explore information at different levels of detail, identify trends and patterns, and support effective business decision-making.

Solution:

- Import the Global Superstore dataset into Microsoft Power BI and load the required tables for analysis.
- Use Power Query to clean and transform the data by creating Fact_Sales and dimension tables such as Dim_Customer, Dim_Product, Dim_Region, and Dim_Date.
- Remove unnecessary columns and duplicate records from the dimension tables to maintain a structured dimensional model.
- Establish suitable one-to-many relationships between the dimension tables and the Fact_Sales table to create a Star Schema.
- Create a Date hierarchy using Year, Quarter, Month, and Date to support detailed time-based analysis.
- Develop DAX measures such as Total Sales, Total Profit, Profit Margin, and Average Order Value for business performance evaluation.
- Create Power BI charts, cards, and slicers to present sales, profit, and order-related information interactively.
- Perform OLAP operations such as Slice and Dice to filter and analyze specific subsets of the dataset.
- Apply hierarchy-based analysis to examine data from higher-level summaries to more detailed levels, supporting effective business decision-making.

0.25

Profit Margin

223.37

Average Order Value

35.38K

Total Profit

2K

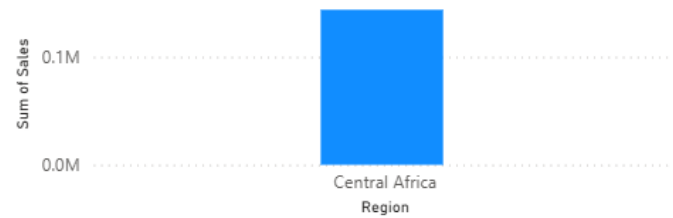
Total Quantity

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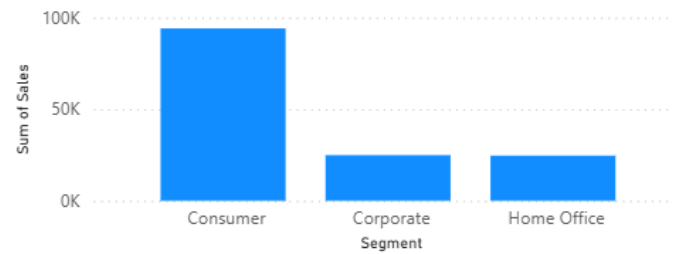
Total Sales



Sum of Sales by Region



Sum of Sales by Segment



Sum of Profit by Category



Region, State, City

- ✓ ☐ Canada
- ^ ☐ Caribbean
 - ✓ ☐ Saint Michael
- ^ ☒ Central Africa
 - ✓ ☒ Benguela
- ✓ ☐ Central America
- ✓ ☐ Central Asia
- ✓ ☐ Central US
- ✓ ☐ Eastern Africa
- ✓ ☐ Eastern Asia
- ✓ ☐ Eastern Europe
- ✓ ☐ Eastern US
- ✓ ☐ North Africa
- ✓ ☐ Northern Europe
- ✓ ☐ Oceania
- ✓ ☐ South America
- ✓ ☐ Southeastern Asia
- ✓ ☐ Southern Africa

1.22M

Total Sales

158.98K

Total Profit

0.13

Profit Margin

217.79

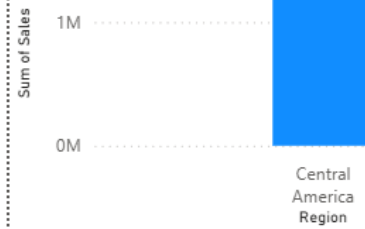
Average Order Value

21K

Total Quantity



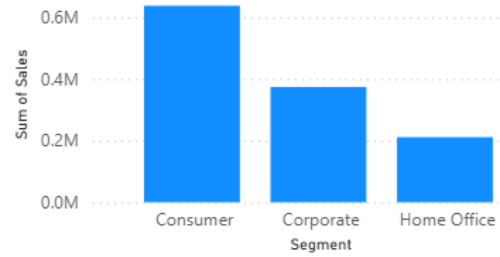
Sum of Sales by Region



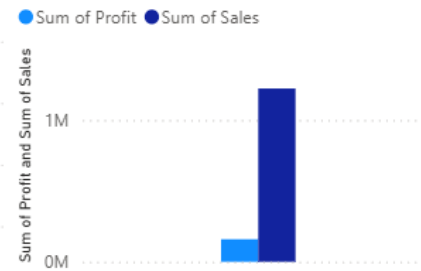
Sum of Profit by Category



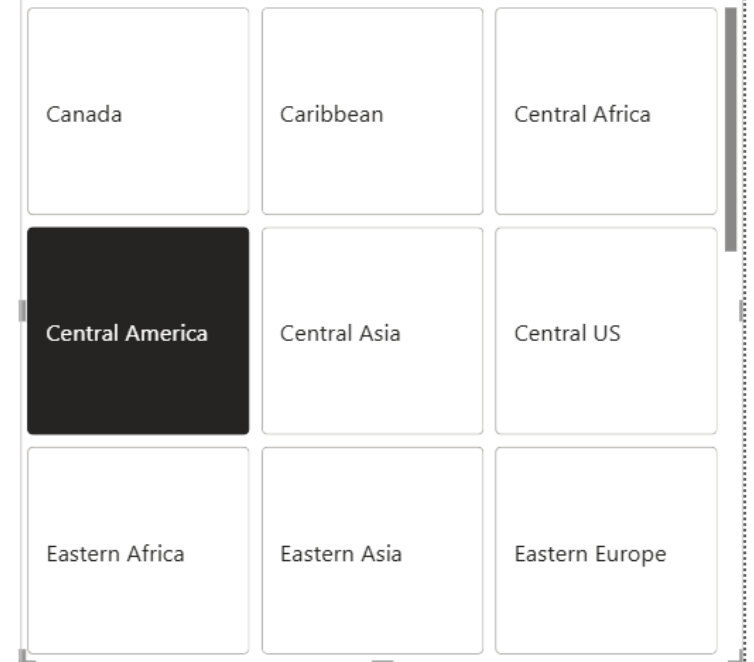
Sum of Sales by Segment



Sum of Profit and Sum of Sales



Region



415.17

Average Order Value

0.16

Profit Margin

74.68K

Total Profit

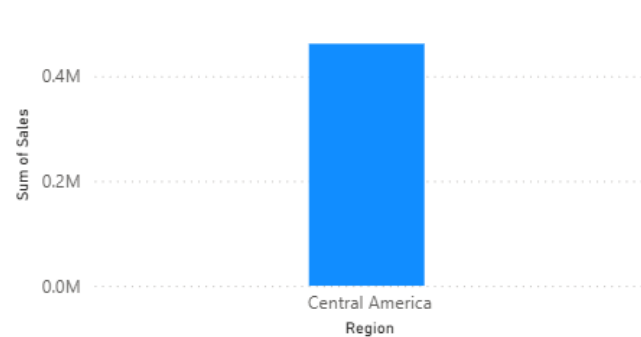
4K

Total Quantity

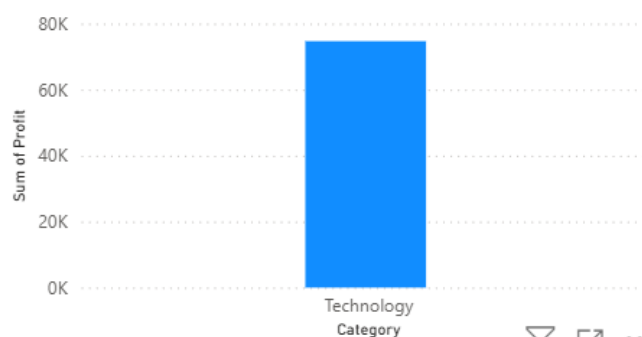
461.67K

Total Sales

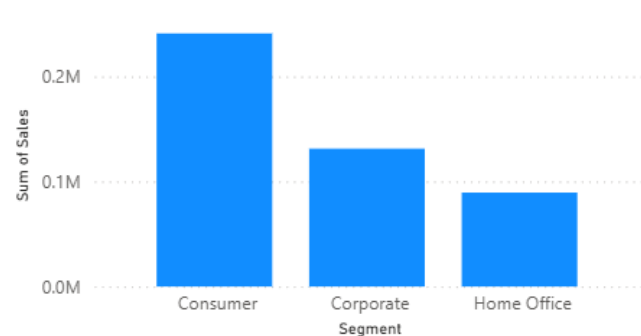
Sum of Sales by Region



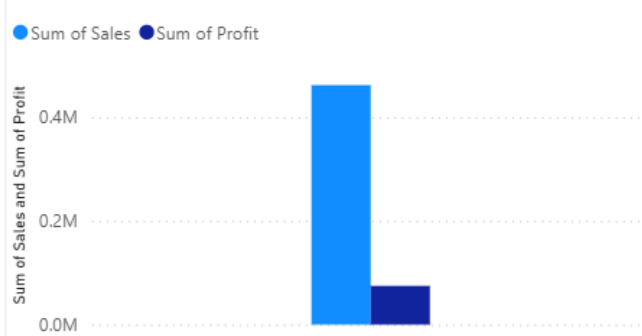
Sum of Profit by Category



Sum of Sales by Segment



Sum of Sales and Sum of Profit



Region

- ☐ Canada
- ☐ Caribbean
- ☐ Central Africa
- ☒ Central America
- ☐ Central Asia
- ☐ Central US
- ☐ Eastern Africa
- ☐ Eastern Asia

Category

- ☐ Furniture
- ☐ Office Supplies
- ☒ Technology

Year

- ☒ (Blank)
- ☐ 2014
- ☐ 2015
- ☐ 2016
- ☐ 2017